DERIVED COLIMITS COMPUTE HOMOTOPY COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. The proof composes four previously displayed engines; transcribed as-is with anchors.

Allowed standard prerequisites (editorial).

- Canonical delta functor on the homotopy category. lemma-derived-canonical-delta-functor (same chapter, carded in the old generation). Supplies the delta-functor engine in the assembly.
- Derived colimit machinery. definition-derived-colimit, lemma-compute-colimit (same chapter, cited in-span; not carded). Define the object being computed and give its representative.
- Direct sums in the homotopy category. lemma-direct-sums (same chapter, cited in-span; not carded). Supplies the coproduct step of the assembly.

Source reference. Original definition 090Z: sources/stacks-derived/pinned-original-source.tex, source lines 10363–10363. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 093W: sources/stacks-derived/pinned-original-source.tex, source lines 10543–10543. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0152: sources/stacks-derived/pinned-original-source.tex, source lines 4182–4182. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A5L: sources/stacks-derived/pinned-original-source.tex, source lines 10517–10517. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $L_n^\bullet$ be a system of complexes of $\mathcal{A}$. Assume colimits over $\mathbf{N}$ exist and are exact in $\mathcal{A}$. Then the termwise colimit $L^\bullet = \operatorname{colim} L_n^\bullet$ is a homotopy colimit of the system in $D(\mathcal{A})$.*

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Proof. We have an exact sequence of complexes

$$0 \to \bigoplus L_n^\bullet \to \bigoplus L_n^\bullet \to L^\bullet \to 0$$

by Lemma 093W. The direct sums are direct sums in $D(\mathcal{A})$ by Lemma 0A5L. Thus the result follows from the definition of derived colimits in Definition 090Z and the fact that a short exact sequence of complexes gives a distinguished triangle (Lemma 0152). □